

Problem 2.1 Stellar population model

Create a simple stellar population model for a star cluster under the assumption of a Salpeter initial mass function (IMF)

$$\log N_s = -1.3 \log M + 2$$

in the mass interval $0.4 - 80 M_\odot$. Use mass bins in logarithmic scale with a step $\Delta \log M = 0.1$.

- a) Compute and plot the initial number of stars as a function of mass.
- b) Calculate and plot the time evolution of the total number of main-sequence stars in the cluster at times $t = 10^6, 10^7, 10^8, 10^9, 10^{10}$ yr. Assume that the main sequence lifetime of a solar mass star is $\tau_{ms} = 10$ Gyr.
- c) Calculate and plot the evolution of the luminosity of the cluster (from main-sequence stars) over the same period of time as assumed in point b).
- d) Derive the total mass to light ratio for the cluster as a function of time. Assume that evolved stars have no mass. Considering also the results of the previous points, what can you say about the contribution of the most and least massive stars in the cluster to the total number of stars and luminosity?
- e) Calculate and plot the evolution of the number of white dwarfs in a cluster of age $t = 10^6, 10^7, 10^8, 10^9, 10^{10}$ yr. Assume all stars with $M < 8 M_\odot$ which reached the end of their main sequence lifetime to be white dwarf.

Problem 2.2 HR- and colour magnitude diagrams

Consider again the star cluster of Problem 1. Assuming no star formation after $t = 0$, construct the corresponding HR-Diagram ($\log L$ vs. T_{eff}) and colour magnitude diagram (M_V vs. B-V) for all stars on the main sequence at $t = 0, 1, 10$ Gyr.

Use the HRdiagram.dat file on Moodle as a reference for the needed quantities. To avoid overlap of all stars of a given mass bin, introduce some Normal distributed scatter in the parameters.